

FORM V MATHEMATICS

Time allowed: 2 hours

Exam date: 14th May 2003

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.

Collection:

- Collect writing booklets in one bundle.
- Start each question in a new writing booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your name, class, and master's initials on each writing booklet:

5P: JNC

5Q: REN

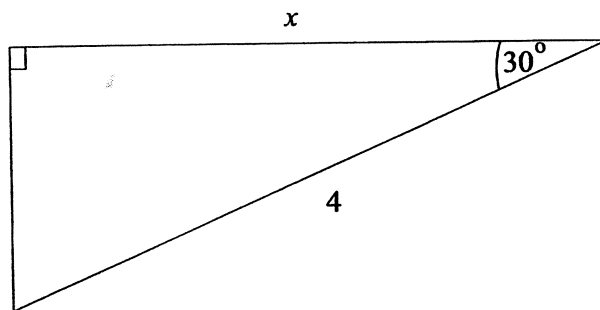
5R: BDD

Checklist:

- Folded A4 writing booklets required — 6 per boy.
- Candidature: 44 boys

QUESTION ONE (Start a new writing booklet)

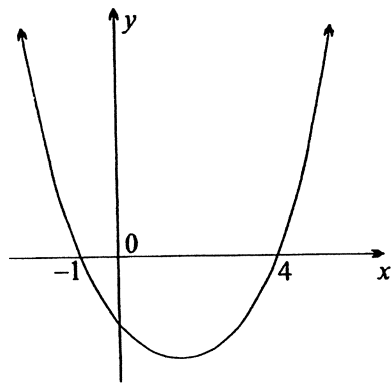
- (a) Solve $5y - 1 = 14$.
- (b) Expand and simplify $(2 + \sqrt{5})^2$.
- (c) Simplify $\sqrt{98}$.
- (d) Write down the exact value of $\tan 120^\circ$.
- (e) Write down the gradient of the line with equation $3x + y + 7 = 0$.
- (f) Express $\frac{6}{\sqrt{3}}$ with a rational denominator.
- (g) Evaluate $|2| - |-7|$.
- (h) Solve the inequation $2x - 1 \leq 5$ and graph the solution on a number line.
- (i) Simplify $4\sqrt{3} + \sqrt{27}$.
- (j) Factorise $2x^2 + 7x - 15$.
- (k) Find the value of $f(2)$ if $f(x) = x^2 + 3x - 1$.
- (l)



In the diagram above, find the exact value of x .

QUESTION TWO (Start a new writing booklet)

- (a) Solve the equation $|x - 4| = 3$.
- (b) Express $\frac{12}{\sqrt{5} - 2}$ with a rational denominator.
- (c) Sketch the graph of $y = x + 1$ on a number plane.
- (d) Find rational numbers x and y such that $x + 3\sqrt{y} = \sqrt{45} - 7$.
- (e)

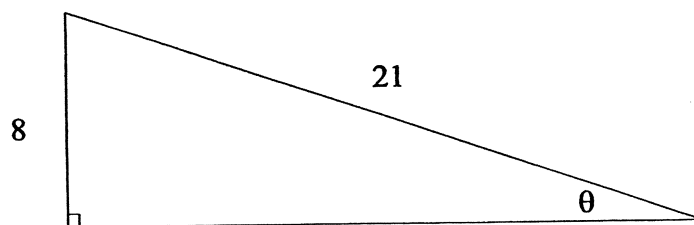


In the diagram above, the graph of $y = x^2 - 3x - 4$ is drawn. Use the graph to solve the inequation $x^2 - 3x - 4 \geq 0$.

- (f) Solve the inequation $|x + 2| < 4$.
- (g) State the domain and range of the function $f(x) = x^2$.
- (h) Shade the region in the number plane in which $y > 3 - x$.

QUESTION THREE (Start a new writing booklet)

(a)



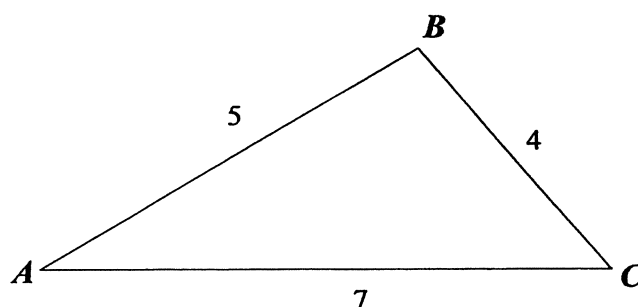
In the diagram above, find the value of θ , correct to the nearest minute.

- (b) Solve $\cos x = -\frac{\sqrt{3}}{2}$, for $0^\circ \leq x \leq 360^\circ$.

(c) Simplify $\frac{\tan \alpha}{\sec \alpha}$.

(d) Given that $\sin \theta = \frac{1}{3}$ and that θ is obtuse, find the exact value of $\tan \theta$.

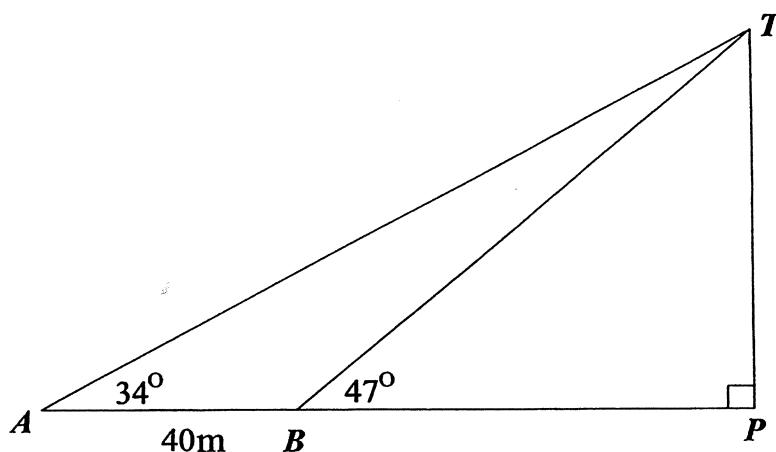
(e) (i)



In the diagram above, use the cosine rule to find the size of $\angle ABC$, correct to the nearest minute.

(ii) Find the area of $\triangle ABC$, correct to one decimal place.

(f)



In the diagram above, the angle of elevation of the top of the tower PT from A is 34° . From a point B 40 metres closer to the tower, the angle of elevation is 47° .

(i) Show that $\angle ATB = 13^\circ$.

(ii) Use the sine rule to find the distance BT , correct to the nearest metre.

(iii) Hence find the height of the tower PT , correct to the nearest metre.

QUESTION FOUR (Start a new writing booklet)

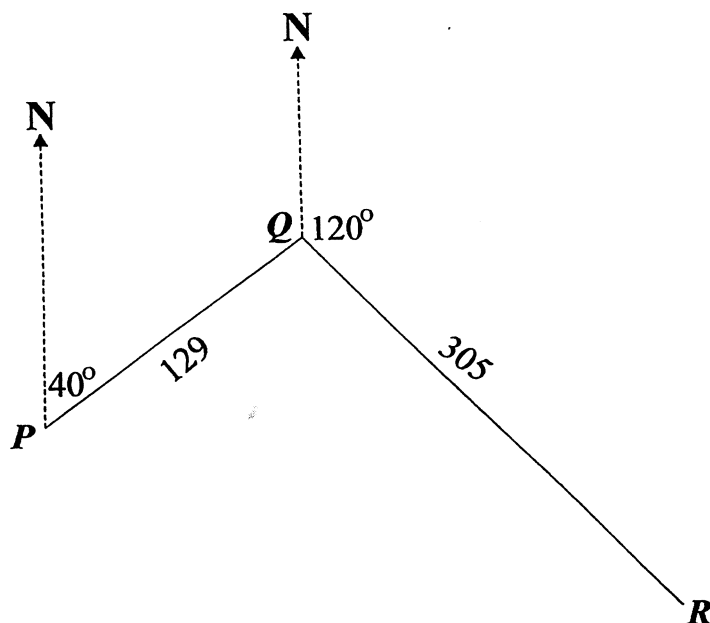
- (a) Write down the gradient of the line that makes an angle of 30° with the positive direction of the x -axis.
- (b) The point A has coordinates $(-2, 1)$ and the point B has coordinates $(-1, -1)$.
 - (i) Find the distance AB .
 - (ii) Find the midpoint of the interval AB .
 - (iii) Find the gradient of the interval AB .
 - (iv) Find the equation of the line that passes through the points A and B .
- (c) Find, in general form, the equation of the line that has a gradient of $-\frac{2}{3}$ and passes through the point $(-2, 1)$.
- (d) Find the equation of the line that passes through the origin and is perpendicular to the line $2x - y + 9 = 0$.
- (e) Show that the point $(-1, 5)$ lies on the line $3x + 2y - 7 = 0$.
- (f) Find the perpendicular distance from the point $(3, 5)$ to the line $2x - y + 4 = 0$.

QUESTION FIVE (Start a new writing booklet)

- (a) Sketch the following functions on SEPARATE number plane graphs showing intercepts, if any, and all other important features:
 - (i) $3x - y + 6 = 0$,
 - (ii) $y = (x - 3)(1 - x)$, (you do not need to give the coordinates of the vertex),
 - (iii) $y = |x + 2|$,
 - (iv) $y = -\frac{1}{x}$.
- (b) Consider the function $f(x) = \sqrt{1 - x^2}$.
 - (i) Sketch the graph of $y = f(x)$.
 - (ii) State the domain of $f(x)$.
 - (iii) Explain why $f(x)$ is an even function. (You may do this by referring to your graph or by giving an algebraic explanation.)
- (c) The equation of a circle is $(x - 3)^2 + (y + 2)^2 = 16$.
 - (i) Write down the coordinates of the centre of the circle and its radius.
 - (ii) Find the perpendicular distance from the line $y = 2x + 3$ to the centre of the circle.
 - (iii) Explain why the line $y = 2x + 3$ does not intersect the circle.

QUESTION SIX (Start a new writing booklet)

- (a) Solve $\sin 2x = -\frac{1}{2}$, for $0^\circ \leq x \leq 180^\circ$.
- (b) (i) Sketch the graphs of $y = |3x - 1|$ and $y = 1 - x$ on the same number plane.
 (ii) Solve $|3x - 1| = 1 - x$.
 (iii) Hence or otherwise, solve the inequation $|3x - 1| > 1 - x$.
- (c) Prove the identity $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = 2 \sec^2 \theta$.
 (Start your proof: $\text{LHS} = \frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta}$)
- (d)



In the diagram above, a ship leaves Port *P* and sails 129 kilometres in a direction of 040° to Port *Q*. From Port *Q* the ship changes direction and sails 305 kilometres in a direction of 120° to Port *R*.

- (i) Explain why $\angle PQR = 100^\circ$.
- (ii) How far is the ship then from Port *P*? (Give your answer correct to the nearest kilometre.)
- (iii) What is the bearing then of Port *P* from the ship? (Give your answer correct to the nearest degree.)

JNC

Question 1

(a) $y = 3$

✓✓

(b) $4 + 4\sqrt{5} + 5 = 9 + 4\sqrt{5}$

✓✓

(c) $7\sqrt{2}$

✓

(d) $-\sqrt{3}$

✓

(e) -3

✓

(f) $\frac{6\sqrt{3}}{3} = 2\sqrt{3}$

✓✓

(g) $2 - 7 = -5$

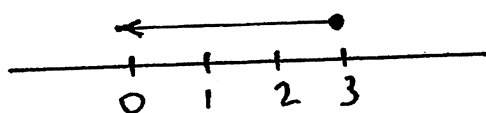
✓

(h) $2x \leq 6$

✓✓

$x \leq 3$

✓



(i) $4\sqrt{3} + 3\sqrt{3}$
 $= 7\sqrt{3}$

✓

✓

(j) $(2x - 3)(x + 5)$

✓

(k) $f(2) = 2^2 + 3 \times 2 - 1$
 $= 9$

✓

(l) $\cos 30^\circ = \frac{x}{4}$

✓

$x = 4 \times \frac{\sqrt{3}}{2}$

$= 2\sqrt{3}$

✓

Question 2

(a) $x - 4 = 3$ or $-(x - 4) = 3$

$\therefore x = 7$ or 1

✓✓

(1 each)

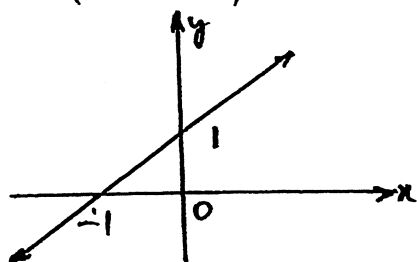
(b) $\frac{12}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2}$

✓

$= 12(\sqrt{5} + 2)$

✓

(c)



✓✓

(d) $x = -7$, and

✓

$3\sqrt{y} = \sqrt{45}$

✓

$3\sqrt{y} = 3\sqrt{5}$

$\therefore y = 5$

✓

(e) $x \leq -1$ or $x \geq 4$

✓✓

(1 each)

(f) $x + 2 < 4$ $-(x + 2) < 4$

$x < 2$ ✓ $x + 2 > -4$

$x > -6$

✓✓

ie $-6 < x < 2$

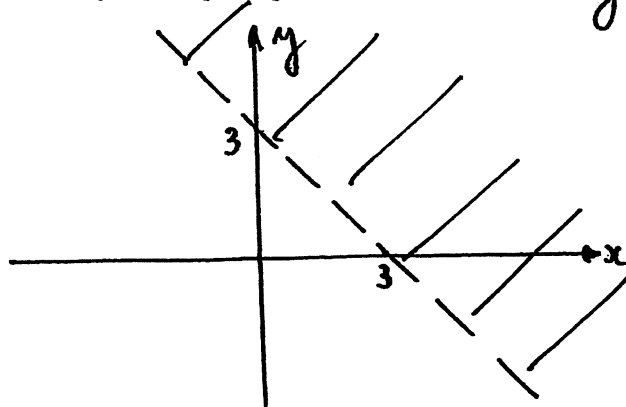
(not needed)

(g) Domain is all real x and range is $y \geq 0$.

✓✓

(1 each)

(h)



✓✓✓

1 line with intercepts

1 dotted

1 shading correct side

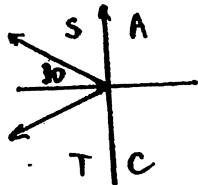
Question 3

(a) $\sin \theta = \frac{8}{21}$ ✓

$\theta = 22^\circ 24'$ ✓

(b) $\cos^2 30^\circ + \sin^2 30^\circ$
 $= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2$
 $= \frac{3}{4} + \frac{1}{4}$
 $= 1$

(b) Related $\angle = 30^\circ$ ✓

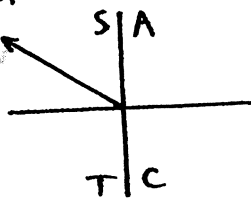
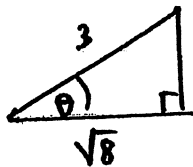


$x = 150^\circ \text{ or } 210^\circ$ ✓✓

(c) $\frac{\frac{\sin \alpha}{\cos \alpha}}{\frac{1}{\cos \alpha}} = \frac{\sin \alpha}{\cos \alpha} \times \frac{\cos \alpha}{1}$

$= \sin \alpha$ ✓✓

(d)



$\tan \theta = -\frac{1}{\sqrt{8}}$ ✓✓

$\left(= -\frac{\sqrt{2}}{4} \right)$

(e) $\cos B = \frac{4^2 + 5^2 - 7^2}{2 \times 4 \times 5}$ ✓
 $= -\frac{1}{5}$ ✓

$\therefore \angle ABC = 101^\circ 32'$ ✓

(ii) Area $\Delta = \frac{1}{2} \times 4 \times 5 \times \sin 101^\circ 32'$ ✓

$= 9.7979...$ ✓

≈ 9.8 ✓

(f)

(i) $\angle ATB + 34 = 47$ (exterior $\angle$ theorem) ✓
 $\therefore \angle ATB = 13^\circ$

(ii) $\frac{BT}{\sin 34} = \frac{40}{\sin 13}$ ✓

$BT = 99.4337...$ ✓

$\approx 99 \text{ m}$ ✓

(iii) $\sin 47 = \frac{PT}{99}$

$\therefore PT = 99 \times \sin 47$ ✓

$= 72.404...$ ✓

$= 72 \text{ m}$ ✓

or 73.

1 for $\sqrt{8}$

1 for -ve

(Wrong formula and correct from there and correct rounding 1/3)

(must be nearest minute for this mark)

(1/2 for wrong formula and no subsequent error)

(don't penalise rounding)

Question 4

(a) $m = \tan 30^\circ$
 $= \frac{1}{\sqrt{3}}$

✓

✓

(b) (i) distance $= \sqrt{1^2 + 2^2}$
 $= \sqrt{5}$

✓✓

(ii) Midpoint $= (-\frac{3}{2}, 0)$

✓✓

(iii) $m = -\frac{2}{1}$
 $= -2$

✓✓

(iv) $y - 1 = -2(x + 2)$
 $y = -2x - 3$

✓

✓

(any form)

(c) $y - 1 = -\frac{2}{3}(x + 2)$

✓

$3y - 3 = -2x - 4$

✓

$2x + 3y + 1 = 0$

✓

(penalise if not in general form)

(d) gradient of line is 2. so

gradient of required line is $-\frac{1}{2}$. ✓

Eqn is:

$y - 0 = -\frac{1}{2}(x - 0)$

✓

$2y = -x$

$x + 2y = 0$

✓

(e) $3x + 2y - 7$

$= 3x - 1 + 2 \times 5 - 7$

$= -3 + 10 - 7$

✓

$= 0$

No point lies on line.

(f) $d = \frac{|6 - 5 + 4|}{\sqrt{2^2 + 1^2}}$

✓

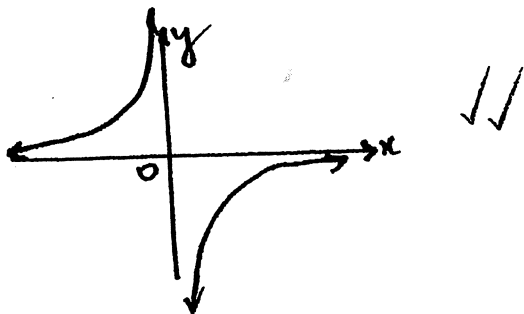
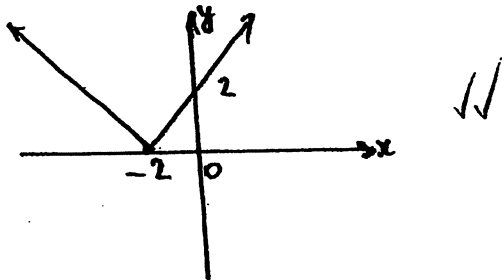
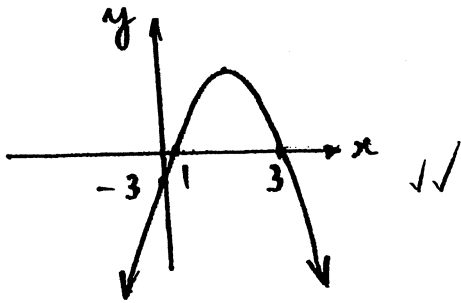
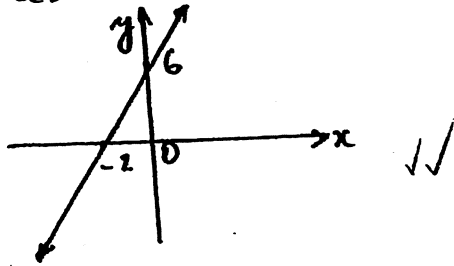
$= \frac{5}{\sqrt{5}} = \sqrt{5}$

✓

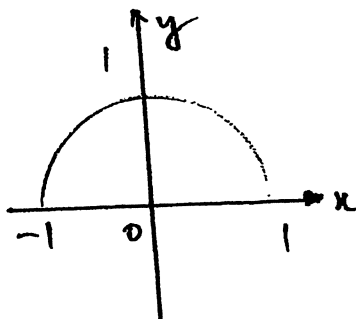
(no need to rationalise)

Question 5

a)



(b)(i)



(ii) domain is $-1 \leq x \leq 1$

(iii) symmetric in y axis, or

$$f(a) = \sqrt{1-a^2}$$

$$f(-a) = \sqrt{1-(-a)^2} = \sqrt{1-a^2} = f(a)$$

(c) (i) centre is $(3, -2)$ ✓

$$r = 4$$

$$(ii) d = \frac{|3 \times 2 + (-2) \times (-1) + 3|}{\sqrt{1^2 + 2^2}} \quad \checkmark$$

$$= \frac{11}{\sqrt{5}}$$

$$= \frac{11\sqrt{5}}{5}$$

iii) $\frac{11\sqrt{5}}{5} \doteq 4.9$ and so the

distance from the line to the centre of the circle is greater than the radius. ✓

Graphs : -1 per error of omission of intercepts

0 marks if wrong shape

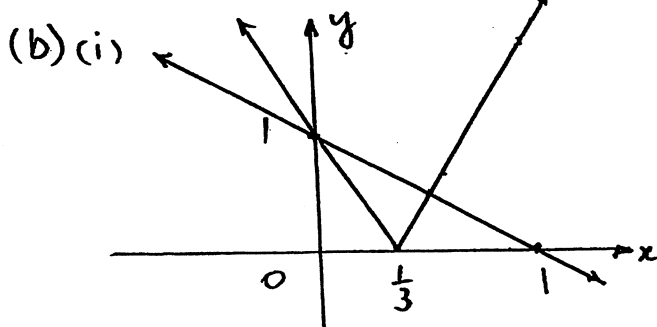
intercepts must be marked

Question 6

(a) Related $\angle = 30^\circ$ ✓

$2x = 210^\circ$ or 330° ✓

$x = 105^\circ$ or 165° ✓



✓ — line

✓✓ — absolute graph

(ii) $|3x-1| = 1-x$

$3x-1 = 1-x$ or $-(3x-1) = 1-x$

$4x = 2$

$2x = 0$

$x = \frac{1}{2}$ ✓✓

$x = 0$ ✓✓

(iii) $x < 0$ or $x > \frac{1}{2}$ ✓

(c) LHS = $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta}$

$= \frac{1-\sin\theta + 1+\sin\theta}{1-\sin^2\theta}$ ✓

$= \frac{2}{\cos^2\theta}$

$= 2\sec^2\theta$ ✓

(d) (i) $\angle PQR = 140^\circ$ (co-interior $\angle$, $\parallel$ lines)

$\angle PQR + 120 + 140 = 360$ ($\angle$ in revolution) ✓✓

$\therefore \angle PQR = 100^\circ$

(ii) $PR^2 = 129^2 + 305^2 - 2 \times 129 \times 305 \times \cos 100^\circ$ ✓

$= 123330.3751 \dots$

≈ 351 ✓

(iii) $\frac{\sin R}{129} = \frac{\sin 100}{351}$

$\sin R = 0.3619 \dots$

$\therefore R \approx 21^\circ$ ✓

So bearing is $270 + (90 - 60 - 21) = 279^\circ$ ✓